

Trace element contents in sphalerite, chalcopyrite and galena in ore deposits from Peru – a meta-analysis

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Abstract. A meta-analysis of trace element contents (LA-ICP-MS data) in sphalerite, chalcopyrite and galena from a variety of Peruvian ore deposits is presented. Compiled data derives from porphyry-related polymetallic (Cordilleran-type and skarn-hosted), Mississippi-Valley type (MVT), and volcanogenic massive sulfide (VMS) deposits. The results show systematic differences in the composition of sphalerite – and to a lesser extent, of chalcopyrite and galena – according to the mineralization style. A better comprehension of distribution patterns may help enhance our ability to target particular sulfide mineralization styles for the benefit of identifying critical raw materials as potential by-products.

1 Introduction

Critical raw materials are essential to the functioning and integrity of a wide range of industrial ecosystems. Several metals and metalloids benefited as by-products from sulfide mineralization – namely In, Ge and Ga – are currently considered as critical raw materials (European Commission 2020) due to their growing usage in both high- and green-tech applications such as solar cells (In, Ge, Ga), optical electronics (In, Ge, Ga), smartphones (In, Ga) amongst others. All three elements rarely form their own mineral phases but are commonly incorporated in the structure of major sulfides such as sphalerite and chalcopyrite (Cook et al. 2009; George et al. 2018).

Through principal component analysis (PCA), recent studies (e.g. Frenzel et al. 2016; Bauer et al. 2019a) have shown that statistically significant differences in the composition (including In, Ge and Ga) of sphalerite from different deposit types exist; also, that sphalerite compositions are distinctively sensitive to the physicochemical characteristics of the mineralizing fluids compared to other major sulfides. Although comparative studies for chalcopyrite and galena from different ore deposit types have also been conducted (e.g. George et al. 2015, 2018), composition trends depending on the deposit type of origin have not been clearly established. Instead, George et al. (2015, 2018) concluded that the compositions of chalcopyrite and galena are primarily influenced by the presence or absence of co-crystallizing sulfides. For example, it is known that chalcopyrite may be the main host of a wide range of trace elements (including In and Ga) in the absence of other co-crystallizing sulfides (e.g. sphalerite and galena; George et al. 2018) and that pre-concentration of trace elements in this mineral may define later concentration in sphalerite along the evolution of

mineralization (Bauer et al. 2021b). To our best knowledge, PCA have not been applied to the composition of chalcopyrite and galena *versus* their deposit type of origin.

Until recent years, little research on the mineralogy and distribution of the so-called high-tech elements had been conducted in Peru. Soler (1987) carried out a geochemical study on Zn concentrates from several Peruvian sulfide deposits and noted a relationship between the contents of Cd, In, Ge, Ga, Ag, Bi, Se, Hg and Sn and the host rock and temperature of ore formation. More recently, systematic in-situ analysis of trace element composition of major sulfides in a variety of ore deposits from the central Andes in Peru has been conducted within the framework of the 107-2018-FONDECYT-BM-IADT-AV research project. In this contribution, we compile recently published and new, unpublished trace element data for sphalerite, chalcopyrite and galena from a variety of Peruvian ore deposits and develop PCAs to test whether significant differences in trace element contents exist.

2 Database and treatment

A total of 1237 LA-ICP-MS spot analyses of ore minerals from magmatic-hydrothermal and hydrothermal ore deposits from Peru (Fig. 1) have been included in the database used in this study. Compiled data stems from porphyry-related polymetallic Cordilleran-type and skarn-hosted deposits (Morococha, Benites et al. 2021a; Ayawilca, Benites et al. 2021b; Zoraida, this study), Mississippi-Valley type deposits (MVT; San Vicente and Shalipayco; Benites et al. 2022), and volcanogenic massive sulfide deposits (VMS; María Teresa, Palma and Perubar; Torró et al. 2022). Raw element contents were log-transformed to ensure approximate normality required for statistical treatment. In order to ensure a statistically robust analysis, data filtering was performed. Analyses with evidence of mineral micro-inclusions (denoted by irregular down-hole ablation spectra) were removed. Elements with >50 % of values either below their respective lower limits of detection (<L.O.D.) or unmeasured contents were also excluded. Nevertheless, in the specific cases of Co (57%) and Se (56%) in sphalerite, this last criterion was not applied because of their distinctively high contents in skarn-hosted polymetallic deposits (see also Benites et al. 2021a). The often-significant proportion of values <L.O.D. is problematic because the ‘standard’ treatment of replacing such values with zero is not possible for log-transformed data. Therefore, an imputation with a fixed value equal to the respective L.O.D. were used for

values <L.O.D. Probability plots and cluster dendrograms with 1-Correlation (1-R), as the dissimilarity measure, were used to complement and validate the PCA results.



Figure 1. Location of the Peruvian ore deposits included in this study.

3 Results

3.1 Sphalerite

PCA results of log-transformed trace element contents in sphalerite are reported in Fig. 2A. A 41.2% of the total variance is captured by principal components (PCs) 1 (mainly Co, In, Cu and Sn) and 2 (mainly Fe and Mn). Biplots including any of the other PCs are not presented since they do not render additional information. The cluster dendrogram (1 – R) shows that In and Cu display a strong positive correlation, as do Se and Co, and Ge and Ag, while Hg, Cd, Pb, Sb and Tl fall into groups with relatively weak positive correlations among the different members (Fig. 2B). Pronounced differences in trace element contents in sphalerite from different ore deposit types occur in the PC1 – PC2 biplot (Fig. 2A). Sphalerite from VMS deposits is particularly enriched in Pb, Sb, Tl, Hg and Cd. MVT deposits host sphalerite enriched in Ge and Ag and, to a lesser extent, Ga. Polymetallic Cordilleran-type deposits show sphalerite enriched mainly in Sn, In, Cu and As and, to some extent, Ga. Sphalerite from skarn-hosted polymetallic deposits are distinctively enriched in Se, Co, Fe, Mn and, to a lesser degree, In.

3.2 Chalcopyrite

PCA results of log-transformed trace element contents in chalcopyrite are reported in Fig. 2C. PC1 (mainly Sb, Cd, Pb, Hg, Tl, Au, Ag, Bi and Zn) and PC2 (mainly Sn and In) capture 52.78% of the variance. Elements

making the largest contributions to the first two PCs accurately reproduce grouping observed in the cluster dendrogram (Fig. 2D). The PC1 – PC2 biplot reveals significant differences in trace element contents in chalcopyrite from different deposit types. Chalcopyrite from VMS deposits is enriched in Au, Tl and Se, and depleted in Sn, In, Ga, Mo and As relative to chalcopyrite from Cordilleran-type deposits. Besides, chalcopyrite from Cordilleran-type deposits is enriched in Zn and Ag. Chalcopyrite from skarn-hosted polymetallic mineralization can be subdivided into two groups: a first group is enriched in Se and Ge (corresponding to the Lower Ombla body), and the second yields moderate contents of Au and Tl (corresponding to the Manto Italia body).

3.3 Galena

PCA results of log-transformed trace element contents in galena are reported in Fig. 2E. 57% of the total variance is captured by PC1 (mainly Cd, Sb, Ag and Bi) and PC2 (mainly Mn and In), which accurately reproduce grouping in the cluster dendrogram (Fig. 2F). Considerable differences in trace element contents in galena from different ore deposit types occur. Galena from VMS deposits is relatively enriched in Cd, Tl, Se and Bi relative to Cordilleran-type deposits. Galena from skarn-hosted polymetallic deposits is particularly rich in Mn and In, as well as Ag and Sb. Galena from MVT deposits is enriched in As and to a lesser extent Se and Ge, and depleted in Sb, Ag and Cd.

4 Discussion

PCA and dendrogram analyses revealed compositional trends for sphalerite, chalcopyrite and galena from different deposit types in Peru. Several authors have cogently argued that the trace element contents in sphalerite reflect the conditions of formation and that systematic composition trends occur depending on the deposit type in which it is deposited (e.g. Cook et al. 2009; Ye et al. 2011; Frenzel et al. 2016; Bauer et al. 2019a). Sphalerite from MVT deposits is typically characterized by a depletion in In, Co and Mn (see also Bauer et al. 2019a; Wei et al. 2021) relative to sphalerite from higher-temperature magmatic-hydrothermal VMS (commonly enriched in Mn and In, and in this study particularly in Cd, up to 9409 ppm; see also Cook et al. 2009; Torró et al. 2022) and porphyry-related polymetallic (commonly enriched in Co, Mn and In; Ye et al. 2011; Bauer et al. 2019b; Benites et al. 2021a,b) deposits. In contrast, sphalerite grains from MVT deposits are enriched in Ge and Ag (see also Benites et al. 2022) and, to some extent, Ga relative to VMS and porphyry-related polymetallic deposits. Therefore, the overall trends in sphalerite observed in this study accord well with previous descriptions that high contents of Ge and Ga are expected in sphalerite from low-temperature deposits with no obvious magmatic connection, while In is enriched in higher-temperature magmatic-hydrothermal deposits (Bauer et al. 2019b). The strong

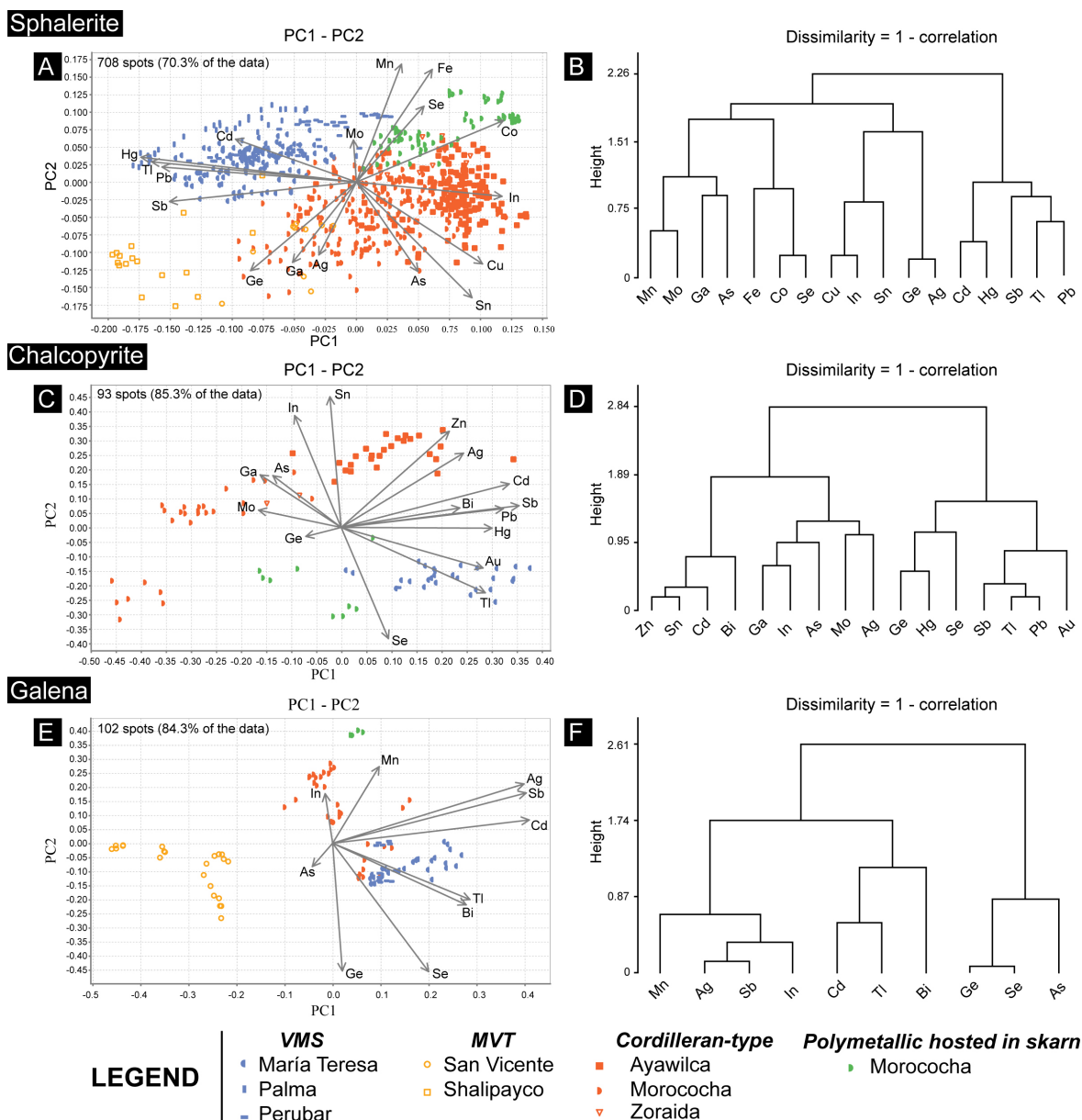


Figure 2. Principal Component Analysis (PCA) biplots (A, C, E) of log-transformed trace element compositions (LA-ICP-MS data) of sphalerite, chalcopyrite and galena from Peruvian ore deposits, and cluster dendrograms (B, D, F) obtained using Ward algorithm with 1 – correlation as the dissimilarity measure. Data sources provided in the main text.

positive correlation between Cu and In in sphalerite (Fig. 2B) is a reflection of dominant $2\text{Zn}^{2+} \leftrightarrow \text{In}^{3+} + \text{Cu}^+$ coupled substitution for the incorporation of this high-tech element to the crystal lattice (Cook et al. 2009). The availability of Cu in mineralizing fluids is hence essential for In enrichment in sphalerite so that Cu-poor ore deposit types (e.g. MVT) are expected to host In-poor sphalerite. The availability of Cu in mineralizing fluids also outlines In enrichment through single deposits/districts as described in Morococha, where In-richest sphalerite in intermediate-sulfidation assemblages in Cordilleran-type polymetallic mineralization occurs in the Cu-Zn, porphyry-proximal zone, whereas sphalerite in distal-to-porphyry Pb-Ag zones are In-poor (see also Torró et al. 2019). Torró et al. (2022) describe progressive In (and Cu) enrichment in sphalerite towards the chalcopyrite-rich base of a

zone-refined massive sulfide lens in the María Teresa VMS deposit. The described strong positive correlation between Ge and Ag in sphalerite (Fig. 2B) is in good agreement with the $3\text{Zn}^{2+} \leftrightarrow \text{Ge}^{4+} + 2\text{Ag}^+$ coupled substitution proposed by Belissont et al. (2014) and Benites et al. (2022).

Chalcopyrite may host a wide range of trace elements including Mn, Co, Zn, Ga, Se, Ag, Cd, In, Sn, Ti (George et al. 2018; Bauer et al. 2019b). Trace element contents showed a moderate variation according to the ore deposits type of origin. We show that Se contents in chalcopyrite from VMS deposits are higher than in chalcopyrite from Cordilleran-type deposits. Likewise, George et al. (2018) concluded that Ga and Sn are prominently incorporated in the chalcopyrite lattice at high temperature and/or pressure conditions. This enrichment is also observed in the PCA

results (Fig. 2C) with both Ga and Sn relatively enriched in chalcopyrite from Cordilleran-type mineralization. The observed positive correlation between Zn and Cd content in chalcopyrite (Fig. 2C-D) was also noted by George et al. (2018). According to George et al. (2015) galena is the main host of Ag, Tl, Bi, Sb and Se. In this study we show that these elements are mainly enriched in galena from VMS deposits and in the case of Sb, also in galena from porphyry-related polymetallic deposits, akin to content patterns described by George et al. (2015) in galena from epithermal deposits.

5 Final remarks

PCA biplots show systematic differences in the composition of sphalerite, chalcopyrite and galena from different mineral deposit types in Peru. Sphalerite from VMS deposits is relatively enriched in Pb, Sb, Tl, Hg and Cd whereas sphalerite from MVT deposits is enriched in Ge, Ag and Ga. Sphalerite from Cordilleran-type deposits is relatively enriched in Sn, In, Cu, Ga and As, and that from skarn-hosted polymetallic deposits is enriched in Se, Co, Fe, Mn and In. The observed trace element trends in the sphalerite support that In ($\pm$ Sn, Co and Mn) is enriched in magmatic-hydrothermal sulfide deposits, while Ge is particularly concentrated in lower-temperature hydrothermal deposits. As for chalcopyrite, it yields a relative enrichment in Au, Tl and Se in VMS deposits, while grains from Cordilleran-type deposits are richer in Sn, In, Ga, Mo, As, Zn and Ag. Galena from VMS deposits is relatively enriched in Tl and Bi.

Regarding high-tech elements, sphalerite is the main host for In (Inter-Quartile Range [IQR] = 166-0.68 ppm, up to 1.7 wt%), Ge (IQR = 3.9-0.15 ppm, up to 1861 ppm), and Ga (IQR = 29-1.5 ppm, up to 2137 ppm) relative to chalcopyrite (IQR = 606-23 ppm, up to 1185 ppm In; IQR = 3.1-0.53 ppm, up to 80 ppm Ge; IQR = 22-2.1 ppm, up to 630 ppm Ga) and galena (IQR = 0.054-0.018 ppm, up to 2.0 ppm In; IQR = 1.4-0.46 ppm, up to 3.1 ppm Ge; Ga <L.O.D for >50 % of the dataset). Indium concentrates chiefly in sphalerite from Cordilleran-type polymetallic deposits (IQR = 484-1.0 ppm, up to 1.7 wt%), whereas Ge is highest in sphalerite from MVT deposits (IQR = 177-45 ppm, up to 1861 ppm). Gallium is mainly enriched in sphalerite from Cordilleran-type (IQR = 48-2.8 ppm, up to 2137 ppm) and MVT (IQR = 12-0.18 ppm, up to 997 ppm) deposits. Chalcopyrite is also an important host for both In (IQR = 858-86.3 ppm, up to 1185 ppm) and Ga (IQR = 29-5.8 ppm, up to 630 ppm) in Cordilleran-type mineralization.

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